

Cross-solver comparative summary

Manufactured verification and Niederer electro-activation benchmark across the five high-order solvers in `cardiacFoam-pc`

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Abstract

This document is an interim cross-report synthesis of the five solver reports currently maintained under `applications/solvers/`: two method-of-manufactured-solutions (MMS) reports that verify the discretisation, and three electro-activation reports that exercise the same discretisation on the Niederer-2012 monodomain benchmark. It is not a substitute for the individual reports; it is meant to answer, in one place, the six practical questions that come up when looking at the runs as a whole: does the method work, what does the benchmark converge to, which discretisation combinations are most cost-effective for that value, what those combinations cost, which combination is the most accurate and which is the most efficient, and what is still missing from the present data. All numbers below are transcribed from the per-solver result tables that exist at the time of writing; no new runs were performed for this summary.

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1 Scope and methods catalogue

The five reports cover the same discretisation toolbox — standard cell-centred finite-volume (FV) diffusion or a Local Regression-Estimator (LRE) high-order reconstruction; cell-centred or LRE-quadrature ionic source; lumped or consistent LRE mass matrix; θ -scheme time integration; and one of three nonlinear coupling strategies (Picard, diagonalIion, JFNK) when the PDE is implicit — applied to two problems:

- A manufactured PDE–ODE system with a known exact solution $(V_m^*, u_1^*, u_2^*, u_3^*)$ and a smooth nonlinear source $I_{\text{ion}}^{\text{MMS}}(V_m, \mathbf{u})$.
- The Niederer-2012 monodomain benchmark with the TNNP ionic model, with t_{act}^{P8} as the diagnostic scalar.

Table 1: Catalogue of the five solver reports.

Solver	Problem	Time scheme	Nonlinear	Dimensions covered
highOrderManufacturedFDAExplicit	NC	forward Euler (explicit)	—	2D ($N = 10 \dots 100$), 3D ($N = 10 \dots 30$)
highOrderManufacturedFDAImplicitJFNK	NC	bE / cN (θ -scheme)	Picard / diagonalIion / JFNK	2D ($N = 10 \dots 100$), 3D ($N = 10 \dots 30$)
highOrderElectroActivationNonExplicitTNNP	NC	forward Euler	—	2D ($\Delta x \in [0.5, 0.05]$ mm), 3D ($\Delta x \in [0.5, 0.1]$ mm)
highOrderElectroActivationNonImplicitTNNP	NC	cN, $\theta = \frac{1}{2}$	single-sweep Picard ($k = 1$)	2D ($\Delta x \in [0.5, 0.05]$ mm, $\Delta t \in [10^{-3}, 10^{-6}]$ s), 3D ($\Delta x \in [0.5, 0.1]$ mm)
highOrderElectroActivationNonImplicitJFNKTNNP	NC	cN, $\theta = \frac{1}{2}$	Picard / diagonalIion / JFNK	2D ($\Delta x \in [0.5, 0.1]$ mm, $\Delta t = 10^{-4}$ s), 3D ($\Delta x = 0.5$ mm)

Throughout this document an LRE pair is written (p_V, p_I) , where the first slot (p_V , for V_m) is the high-order level used for the diffusion operator (and, when consistent mass is on, for the mass matrix) and the second slot (p_I , for I_{ion}) is the level used for the ionic source and ODE quadrature. N0 means the standard FV / cell-centre operator. NC marks any cell where the corresponding run has not yet been done.

2 Convergence verification (MMS)

2.1 What the LRE level is supposed to deliver

The two MMS reports state the same theoretical-accuracy ladder: N0 and p1 target second-order spatial accuracy, p2 third-order, and p3 fourth-order. The cell-centred FV operator itself is second-order, so the cell-centred error column can only saturate at ~ 2 ; the formal LRE order can only be seen in the *reconstructed* (Gauss-quadrature) error columns. The fitted order is also the minimum of the spatial order and the temporal / nonlinear-tolerance order; an aggressive LRE polynomial does not recover its formal order unless Δt , the mass matrix, and the nonlinear residual all permit it.

2.2 Explicit MMS — observed orders

From `highOrderManufacturedFDAExplicit/report/tables/explicit_orders_2D.tex` and `explicit_orders_3D.tex`, fitted on $N = 10 \dots 100$ (2D) and $N = 10 \dots 30$ (3D), at the finest $\Delta t = 10^{-6}$ s:

Table 2: Explicit MMS: observed convergence orders at $\Delta t = 10^{-6}$ s. Cell / G2 = cell-centred / reconstructed Gauss-quadrature error. Targets are written as V_m/I_{ion} (LRE polynomial order).

$(\mathbf{p}_V, \mathbf{p}_I)$	target	V_m cell (2D)	V_m^G (2D)	V_m cell (3D)	V_m^G (3D)
(N0,N0)	2/2	1.997	1.997	1.979	1.979
(p1,p1)	2/2	1.912	2.147	2.008	2.063
(p2,p2)	3/3	2.244	2.762	2.021	3.289
(p3,p3)	4/4	2.369	3.091	3.108	3.869

Reading the table: N0/N0 and p1/p1 are clean second-order in both columns; p2/p2 reaches its target third order in the reconstructed column in 3D (3.289) and is close in 2D (2.762); p3/p3 hits 3.869 in 3D — formal fourth order is approached but not fully recovered with explicit forward-Euler at $\Delta t = 10^{-6}$ s. The cell-centred V_m column saturates at ~ 2 in 2D as expected.

2.3 Implicit MMS — where formal fourth order is recovered

The implicit JFNK MMS report sweeps the same LRE pairs with both time schemes (bE/cN), both mass matrices (lumped/consistent), and all three nonlinear methods. The cleanest case for fourth order is p3/p2 with crankNicolson, consistent mass, and $\Delta t = 10^{-3}$ s (tables/results_2D*_dt_1p0em03.tex, line 65):

Table 3: Implicit MMS 2D, p3/p2, $\Delta t = 10^{-3}$ s. Orders fitted over $N = 10 \dots 100$. Cost is the t_Σ / RSS_Σ row sum from the same table.

Configuration	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G
bE / consistent	2.434	3.214	2.000	3.534	2.000	3.532
cN / consistent	3.539	3.993	1.999	4.003	1.997	4.010
cN / lumped	2.369	3.091	1.999	3.441	1.996	3.435

Three concrete observations from those three rows:

- **Backward Euler caps to first order in time**, which here collapses the reconstructed V_m fit from $3.993 \rightarrow 3.214$; the state-reconstructed columns survive because their dominant error component is spatial, not temporal.
- **Lumped mass caps the reconstructed V_m order** ($3.993 \rightarrow 3.091$); the consistent LRE mass matrix is what lets the reconstruction respect its own quadrature.
- **Crank–Nicolson with consistent mass is the only combination that recovers the formal fourth-order target** for both V_m^G and the reconstructed state errors (3.993, 4.003, 4.010).

Comparing the three nonlinear methods on the very same p3/p2, cN, consistent, $\Delta t = 10^{-3}$ s row:

Nonlinear method	V_m	V_m^G	t_Σ [s]	RSS_Σ [MB]
Picard	3.538	3.992	1730.1	19265.3
diagonalIion	3.538	3.992	1661.8	19310.4
JFNK	3.539	3.993	798.3	18963.3

The three methods produce identical orders to three decimals — as expected, since the MMS source is smooth and the nonlinear coupling error is small. They differ purely in cost: JFNK is $\sim 2.2\times$ cheaper than Picard / diagonalIion in this MMS test. *This is the opposite of the electro-activation picture below*, and the difference is explained in §4.

2.4 Bottom line — does the method work?

Yes. The MMS evidence is consistent: the formal LRE polynomial order is recovered whenever the rest of the discretisation does not block it (Crank–Nicolson, consistent mass, Δt small enough, nonlinear residual converged). The MMS does not stress-test the non-smoothness of an activation front or the stiffness of the TNNP ionic model; those stress tests are the role of the Niederer reports.

3 Electro-activation benchmark (Niederer)

This is the same monodomain PDE + TNNP-2004 ionic model in three geometries: a 2D code-development reduction (20×8 mm, single cell in z , planar fibres; *not* a published Niederer geometry) and the published 3D Niederer slab ($20 \times 3 \times 7$ mm, anisotropic fibre rotation, $40 \times 6 \times 14$ cells at $\Delta x = 0.5$ mm). The scalar diagnostic is the P8 activation time t_{act}^{P8} [ms]: the time at which a designated benchmark point at the far end of the slab first crosses $V_m = V_{\text{thr}}$.

3.1 2D internal-consistency floor

Neither the 2D explicit nor the 2D implicit report can compare t_{act}^{P8} against an external reference (the 2D reduction is not in the published benchmark literature). Instead each report reports an *internal-consistency floor*: the band in which the high-order LRE pairs all agree at the finest mesh / smallest Δt .

Table 4: 2D internal-consistency floor for t_{act}^{P8} across the three electro-activation reports.

Solver	Floor for t_{act}^{P8} [ms]	Where it is reached	Reached by
...FoamExplicit	[44.5, 46.0]	$\Delta x \leq 0.1$ mm	(N0,p2), (N0,p3), (p1,p2), (p1,p3), (p3,p3)
...FoamImplicit	[45.04, 45.80]	$\Delta x = 0.1$ mm, $\Delta t = 10^{-6}$ s	every HO pair we ran at the finest column
...FoamImplicitJfNK	(no floor yet)	2D finest mesh available is $\Delta x = 0.1$ mm and only for Picard	(N0,N0)=42.76, (N0,p1)=41.84; HO pairs were not run at $\Delta x = 0.1$ mm yet

The first two solvers agree to within ~ 0.5 ms on a floor of $t_{\text{act}}^{P8} \approx 45\text{--}46$ ms for this 2D reduction. The JfNK report has not yet been pushed deep enough into the high-order / fine-mesh corner to corroborate that floor; the only $\Delta x = 0.1$ mm entries it has are low-order Picard rows that sit slightly below it ((N0,N0) drops to 42.76 ms when Δt goes from 10^{-5} to 10^{-4} s, which is the single-Picard linearisation effect discussed in §4).

3.2 3D current best estimate

The 3D data are much sparser. Tabulating what each electro-activation report has at $\Delta x = 0.5$ mm (the only mesh that exists in all three):

Table 5: 3D t_{act}^{P8} [ms] at $\Delta x = 0.5$ mm across the three electro-activation reports. NC marks LRE pairs that were not yet run with that solver.

Pair	Explicit ($\Delta t_{\text{req}} = 5 \cdot 10^{-5}$ s)	Implicit ($\Delta t = 5 \cdot 10^{-5}$ s)	JfNK-Picard ($\Delta t = 10^{-4}$ s)	JfNK-diagLion ($\Delta t = 10^{-4}$ s)
(N0,N0)	141.5	142.7	136.7	136.6
(N0,p1)	36.4	38.0	33.5	33.4
(N0,p3)	NC	NC	30.6	30.5
(p3,N0)	NC	NC	149.1	149.1
(p3,p1)	NC	NC	32.4	32.4
(p3,p3)	32.6	34.5	29.6	NC

The qualitative pattern is the same in all three reports: turning on LRE on the ionic source alone collapses the activation time from ~ 140 ms (low-order) to ~ 30 – 38 ms; the matched HO pair adds another ~ 2 – 5 ms drop. The three solvers agree within ~ 0 – 8 ms across all rows, with the dominant offset coming from Δt ($5 \cdot 10^{-5}$ vs 10^{-4} s) and from the single-Picard linearisation in the non-JFNK implicit branch.

The 3D problem does not yet have enough mesh refinement to claim a converged t_{act}^{P8} . The high-order pairs cluster in $[29.6, 34.5]$ ms at $\Delta x = 0.5$ mm, but the only finer 3D mesh that exists is for the low-order pair (N0,N0) (implicit: 55.6 ms at 0.2 mm, 47.3 ms at 0.1 mm). Until at least one HO pair is rerun at $\Delta x \leq 0.25$ mm, the 3D internal-consistency floor remains open.

3.3 Cross-method agreement on the same configuration

The implicit-JFNK report contains the only side-by-side comparison of Picard, diagonalLion, and JFNK on the same electro-activation case. At $\Delta x = 0.5$ mm, $\Delta t = 10^{-4}$ s:

Pair	Picard [ms]	diagonalLion [ms]	JFNK [ms]
(N0,N0)	153.32	153.16	153.13
(N0,p3)	32.46	32.37	32.36
(p3,p3)	33.09	33.00	33.00

The three nonlinear methods agree to within 0.1–0.2 ms on every row available. In other words, on the Niederer benchmark the *choice of nonlinear method is essentially accuracy-free* at $\Delta t = 10^{-4}$ s; it is a cost decision (see §5).

4 Most cost-effective combinations

4.1 High-order on the source alone is the dominant accuracy lever

The single largest accuracy gain per unit of extra cost comes from turning on LRE quadrature for the ionic source while leaving the diffusion at the standard FV operator: the (N0,p2) or (N0,p3) pair. Concrete example from the implicit 2D table (`...FoamImplicit/report/results_tables.tex`, $\Delta t = 10^{-5}$ s):

Pair	$\Delta x = 0.5$	0.25	0.125	0.1	0.05 [mm]
(N0,N0) t_{act}^{P8} [ms]	157.69	67.69	50.65	48.55	46.10
(N0,p2) t_{act}^{P8} [ms]	35.26	43.20	45.15	45.42	—
(N0,p2)/(N0,N0) wall-clock ratio	2.0	3.0	5.5	4.5	—

Reading row 1: the low-order pair only enters the floor band at the two finest meshes. Reading row 2: (N0,p2) reaches a value within ~ 2 ms of the floor already at $\Delta x = 0.5$ mm and is inside the band at $\Delta x = 0.125$ mm. The cost ratio (row 3) is between $\sim 2\times$ and $\sim 5\times$ for the same

target accuracy — in other words, far cheaper than the order-of-magnitude mesh refinement that the (N0,N0) branch would need to match it.

The same pattern holds in the explicit report ((N0,p2) reaches $34.7 \rightarrow 44.95$ ms over the mesh sweep) and in the implicit-JFNK report ((N0,p3) reaches $32.4 \rightarrow$ finer-mesh values within the band).

4.2 LRE on the diffusion only is the worst-of-both-worlds

The (pk, N0) branch — high-order LRE Laplacian on top of a piecewise-constant ionic source — is documented in the explicit report as numerically pathological on coarse meshes. The Niederer explicit 2D table shows (p1,N0) at $\Delta x = 0.5$ mm giving $t_{\text{act}}^{P8} = 14.25$ ms (a spurious very-early crossing) and (p2,N0) giving 0.00 ms (the front never crossed in the simulated window). Same pattern in the implicit-JFNK 2D table: (p3,N0) at $\Delta x = 0.5$ mm gives 180.4–180.5 ms across all three nonlinear methods, $\sim 20\%$ worse than the (N0,N0) baseline at the same mesh. This branch should be treated as a diagnostic, not a production setting.

4.3 The best combination versus the most efficient combination

Putting the two together:

- **Best combination** (highest fidelity reference). A matched high-order pair (p3/p3 or p3/p2) with Crank–Nicolson, consistent mass, the finest mesh in the table, and $\Delta t = 10^{-6}$ s on the implicit solver. This is the combination that *defines* the internal-consistency floor in 2D (45.0–45.8 ms). The single-Picard cost in this corner is ~ 700 –900 min wall-clock on the finest 2D mesh per case (...FoamImplicit/results_tables.tex wall-clock block).
- **Most efficient combination** (highest accuracy per unit cost). (N0,p2) or (N0,p3) on a coarse mesh, with Picard as the nonlinear method when the implicit branch is needed — or with explicit forward-Euler when the CFL clamp at $\Delta x = 0.5$ mm ($\Delta t_{\text{CFL}} \approx 8.7 \cdot 10^{-5}$ s in 3D) is acceptable. Picard and diagonalIion agree on t_{act}^{P8} in both 2D and 3D, and Picard is typically ~ 10 –20% faster than diagonalIion at this accuracy level on the Niederer problem.

JFNK on the electro-activation problem is *not* the most efficient choice, despite being so on the MMS problem. The reason is that the dominant cost of a JFNK iteration on the electro-activation problem is the per-Krylov-vector TNNP ODE re-solve (RKF45, 17 states, adaptive sub-stepping); each GMRES matvec re-integrates the ionic ODE over $[t^n, t^{n+1}]$ on every cell or quadrature point, which is much more expensive than the cheap manufactured source of the MMS test. Concrete penalty from the implicit-JFNK 2D wall-clock table (...FoamImplicitJfNK/.../solver_report.tex Table 6) at $\Delta x = 0.125$ mm, (N0,p3): Picard 402 min vs JFNK 3407 min ($\sim 8.5\times$ penalty; diagonalIion sits between them at 533 min).

5 Computational cost cross-table

The following table lists four landmark configurations, one per row, and reports their wall-clock cost (in minutes) and peak resident set size (in MiB) on the cheapest available solver / nonlinear method. The numbers come directly from the per-solver result tables.

Table 6: Cost cross-table for four landmark configurations. Time is elapsed wall-clock in minutes (rounded). NA = configuration not applicable to that solver.

Configuration	Solver / method	2D time	2D RSS	3D time
Cheapest baseline:	Explicit	1 min	148 MiB	3 min
(N0,N0), $\Delta x = 0.5$ mm	Implicit / Picard, $\Delta t = 10^{-3}$	1 min	148 MiB	4 min ($\Delta t = 5e-5$)

Configuration	Solver / method	2D time	2D RSS	3D time
	JfNK / Picard, $\Delta t = 10^{-4}$	15 min	148 MiB	80 min
Cheapest accurate:	Explicit (CFL-clamped)	1 min	152 MiB	11 min ($\Delta x = 0.5$, Δt_{CFL})
(N0,p2) or (N0,p3), $\Delta x = 0.5$ mm	Implicit / Picard, $\Delta t = 10^{-5}$	2–3 min	155 MiB	14 min ($\Delta x = 0.5$, $\Delta t = 5e-5$)
	JfNK / Picard, $\Delta t = 10^{-4}$	23 min	159 MiB	168 min
Mid-range converged:	Explicit	42 min	234 MiB	NA
(N0,p2), $\Delta x = 0.125$ mm	Implicit / Picard, $\Delta t = 10^{-5}$	44 min	269 MiB	NA
	JfNK / Picard, $\Delta t = 10^{-4}$	402 min	288 MiB	NA
Reference run (defines floor):	Explicit ($\Delta x = 0.05$ mm)	1167 min	961 MiB	NA
(p2,p2), finest mesh	Implicit / Picard, $\Delta x = 0.05$, $\Delta t = 10^{-3}$	246 min	1429 MiB	NA
	Implicit / Picard, $\Delta x = 0.1$, $\Delta t = 10^{-6}$	730 min	466 MiB	NA

The two general lessons that fall out of this table are:

- **Crossing into a finer mesh costs 10–30× more than upgrading the LRE pair on a coarse mesh;** e.g. going from (N0,p2) at $\Delta x = 0.5$ mm (2 min) to (N0,p2) at $\Delta x = 0.125$ mm (44 min) is a $\sim 20\times$ cost step, and buys only ~ 10 ms of accuracy (t_{act}^{P8} moves from 35.3 to 45.2 ms, i.e. from below to inside the floor band).
- **The JFNK branch on the electro-activation problem multiplies wall-clock cost by $\sim 5\text{--}8\times$ vs Picard at the same accuracy;** see for instance the (N0,p3)/ $\Delta x = 0.125$ mm row of the JFNK 2D table cited in §4. Memory cost is essentially method-independent.

6 Gaps and pending runs

The largest NC blocks in the current data, listed in the order in which closing them most increases the value of the reports:

1. **JFNK 3D sweep has not been run at all.** The implicit-JFNK 3D table has Picard and diagonalIion columns but no JFNK column. Filling at least the six 2D-row configurations at $\Delta x = 0.5$ mm closes the cross-method comparison in 3D.
2. **One finer 3D mesh on a high-order pair.** The implicit 3D table only has (N0,N0) at $\Delta x \in \{0.5, 0.2, 0.1\}$ mm; no high-order pair has been run below $\Delta x = 0.5$ mm in 3D. A single run of (N0,p2) or (p3,p3) at $\Delta x = 0.25$ mm or 0.2 mm (cheapest on explicit forward-Euler at $\Delta t = 5 \cdot 10^{-6}$ s, or on Picard at $\Delta t = 10^{-4}$ s) would let us define the 3D internal-consistency floor.
3. **Missing (p3,p3) diagonalIion 3D row** in the implicit-JFNK 3D table.
4. **Finest-mesh entries for (N0,p2) and (N0,p3) in the explicit 2D table** at $\Delta x = 0.05$ mm are NC; these are the runs that would let us claim the explicit 2D floor *also* from the most cost-effective LRE pairs, not only from the matched high-order pairs.
5. **Several Picard-only entries at $\Delta t = 10^{-6}$ s, $\Delta x = 0.05$ mm** in the implicit (non-JFNK) 2D table are NC (mostly the p3 rows). These are the heaviest cases in the entire 2D study; they are not blocking conclusions, only documentation.

6. **JFNK 2D fine-mesh rows.** JFNK 2D currently stops at $\Delta x = 0.125$ mm; only Picard reaches $\Delta x = 0.1$ mm. Two or three JFNK runs at the same mesh as the existing Picard rows would close the JFNK 2D internal-consistency picture.

Items (1) and (2) are the two runs that most change the qualitative conclusion of this summary; everything else is documentation.

7 Bottom-line conclusions

Answering, in order, the six questions that prompted this document:

1. **Do the methods work?** Yes. The MMS reports recover the formal LRE polynomial order on every clean combination (Crank–Nicolson, consistent mass, Δt small enough). The three Niederer-electro-activation reports agree with each other in the overlap regions; cross-method differences are explained by their respective discretisation choices, not by a bug.
2. **What does the electro-activation problem converge to?** **2D:** $t_{\text{act}}^{P8} \approx 45\text{--}46$ ms (internal-consistency floor shared by the explicit and the single-Picard implicit reports). The JFNK report has not yet run deep enough to corroborate this floor from a different time-step / nonlinear path. **3D:** the answer is currently *open*. The high-order pairs cluster in 29.6–34.5 ms at $\Delta x = 0.5$ mm but the 3D mesh sweep is too shallow to define a converged value.
3. **Which combinations are the most cost-effective for that value?** (N0,p2) or (N0,p3) — LRE quadrature on the ionic source alone, with the standard FV diffusion. They reach the internal-consistency band already at $\Delta x = 0.125$ mm and cost $\sim 5\times$ less than the matched high-order pairs at the same mesh.
4. **What do those combinations cost?** On the implicit (single-Picard) 2D table at $\Delta t = 10^{-5}$ s: ~ 2 min at $\Delta x = 0.5$ mm, ~ 44 min at $\Delta x = 0.125$ mm, ~ 243 min at $\Delta x = 0.05$ mm for (N0,p2); peak RSS stays under 1 GiB. JFNK multiplies the wall-clock by $\sim 5\text{--}8\times$ for the same configuration on the electro-activation problem.
5. **Which is the best combination?** The matched high-order pair p3/p3 (or p3/p2) with cN + consistent mass + finest mesh + smallest Δt . This is the combination that *defines* the floor; it is also the single most expensive cell of the entire study.
6. **Which is the most efficient combination?** (N0,p2) or (N0,p3) on a coarse mesh, with Picard when the implicit branch is needed. On the electro-activation problem the JFNK branch is *not* efficient (the per-Krylov ODE re-solve of the TNNP cell model dominates); this is the qualitative reverse of the MMS picture in §2, where JFNK is the cheapest of the three.

A note on Crank–Nicolson order. The implicit (non-JFNK) electro-activation report documents that with a *single* Picard sweep per time step, the formal $\theta = \frac{1}{2}$ second-order temporal behaviour is not recovered (t_{act}^{P8} on the finest mesh halves between $\Delta t = 10^{-3}$ and 10^{-4} s, rather than quartering as the asymptote would predict). This is a linearisation-error signature, not a Crank–Nicolson failure; the implicit-JFNK solver fixes it by genuinely solving the coupled nonlinear residual. The implicit report’s stated next step — enabling a tolerance-based stop on the Picard residual with $k \geq 2$ sweeps — is the cheapest way to recover CN second order without paying the JFNK ODE-re-solve cost.